UML Diagrams & System Architecture

Enterprise Clean Architecture, State Machine Specifications, Sequence Flows & Entity Data Models • Android Jetpack & Hardware Keystore

1. Architectural Overview & Clean Architecture

Umkhuseli follows the official Android Enterprise Clean Architecture standard, structured into strictly decoupled layers: **UI (Presentation)**, **Domain / ViewModel**, **Dependency Injection (AppContainer)**, and **Data / Hardware Services**. To ensure zero overhead, deterministic initialization, and instant 0ms memory scrubbing, the application uses a **Manual DI Container** (AppContainer) instead of heavyweight reflection-based injection frameworks.

Architecture Layer	Primary Components	Key Responsibilities & Technologies
Presentation / UI Layer	MainActivity LauncherFacadeScreen SafeHavenScreen SafetyTriggerOverlay	Declarative Jetpack Compose UI; renders African proverbs facade, biometric Safe Haven, and panic overlay state machine. Zero XML.
Domain & ViewModel Layer	SafetyTriggerViewModel SankofaViewModel ViewModelFactory	Manages UI StateFlow streams; orchestrates 3-second SOS grace period countdown, duress PIN flows, and incident sorting/filtering.
Dependency Injection	AppContainer SankofaApplication	Singleton manual DI container; provides thread-safe access to Room DB, AndroidKeyStore crypto helpers, and dispatch services.
Data & Persistence Layer	AppDatabase IncidentDao GuardianDao UserProfileDao	Room SQLite persistence; stores encrypted incident narratives, trusted ICE guardians, and statutory Section 4 complainant profile.
Hardware & Security Layer	VaultEncryptionHelper EmergencyService EmergencyDispatchHelper	AndroidKeyStore 256-bit AES-GCM StrongBox keys, BiometricPrompt, LocationManager GPS queries, SmsManager broadcasts, and Vibrator.

2. UML State Machine: Emergency SOS Signal Lifecycle

The emergency dispatch system operates as a finite state machine implemented via SafetySignalState sealed interface. State transitions guarantee tactile feedback on countdown ticks and atomic persistence to Room DB on dispatch:

Source State	Transition Event	Target State	System Actions & Side Effects
[Idle]	armQuickAccessSos(3)	**[Armed(sec=3)]**	Launches countdown coroutine; triggers initial haptic vibration pulse.
[Armed(sec>0)]	1-second tick (sec ≥ 1)	**[Armed(sec=sec-1)]**	Updates countdown state; pulses vibrator for tactile confirmation.
[Armed]	cancelArmedSos()	**[Cancelled]**	Cancels coroutine; prevents SMS dispatch; returns to idle upon user acknowledgement.
[Armed]	Countdown reaches 0	**[Dispatched]**	Fetches GPS coordinates; persists IncidentEntity; broadcasts 72-hr beacon SMS to 5 guardians.

[Idle / Armed]	triggerInstantSos()	**[Dispatched]**	Bypasses countdown; immediately persists emergency incident and fires guardian SMS broadcast.

3. UML Sequence Flows

Sequence 1: Safe Haven Entry & Biometric Decryption

Actor: Survivor • **Components:** LauncherFacade → SankofaViewModel → BiometricPrompt → VaultEncryptionHelper → Room DB

1. Survivor inputs PIN 1234 or triple-taps Sankofa emblem on home launcher.
2. LauncherFacade notifies SankofaViewModel without rendering audio/visual alert.
3. BiometricPrompt prompts biometric challenge; user provides verified fingerprint.
4. VaultEncryptionHelper accesses sankofa_vault_master_key from AndroidKeyStore.
5. IncidentDao.getAllIncidents() queries encrypted ciphertexts from Room SQLite.
6. VaultEncryptionHelper.decrypt() initializes AES-256-GCM cipher with record's 12-byte IV.
7. Decrypted victim narratives and J88 clinical notes are rendered inside SafeHavenScreen.

Sequence 2: 0-Millisecond Panic Camouflage Scrub

Actor: Survivor / Approaching Intruder • **Components:** Hardware Back Key → BackHandler → SankofaViewModel → RAM → UI

1. Intruder approaches; survivor presses hardware Back button or Top Shield Bar.
2. Android BackHandler intercepts event synchronously (0 ms latency).
3. SankofaViewModel.resetToCamouflage() zeroes plaintext memory references and resets UI StateFlow to Idle.
4. Compose recomposes directly to LauncherFacadeScreen (African proverbs display).
5. Android task thumbnail is sanitized, leaving zero visual traces in the Recent Apps switcher.

4. UML Class Diagram: Data Entities & DAO Relationships

Entity / Class	Key Fields & Types	Statutory / Legal Purpose & Methods
IncidentEntity **(Room Table: *incidents*)**	id: Long timestamp: Long incidentType: String severityLevel: Int (1-5) perpetrator: String policeObNumber: String description: String (ENC) injuriesOrImpact: String (ENC)	Court-admissible incident log compliant with Domestic Violence Act Form 2 affidavit requirements. Sensitive narrative fields encrypted via AES-256-GCM.
GuardianEntity **(Room Table: *guardians*)**	id: Long name: String surname: String mobileNumber1: String hasWhatsAppAccess: Boolean isPrimary: Boolean instagramHandle: String tiktokHandle: String	Trusted Emergency ICE Guardian registry. Stores primary/secondary contacts and social handles for multi-channel live beacon alerts.
UserProfileEntity **(Room Table: *user_profile*)**	id: Long = 1L fullLegalNames: String idOrPassportNumber: String safeContactNumber: String safeServiceAddress: String isAddressWithheld: Boolean digitalSignatureBase64: String duressPin: String = '9999'	Section 4 Domestic Violence Act Complainant Profile. Stores statutory legal particulars and Section 4(2) residential address withholding toggle for court protection.